

Original condition: full investigation report

Everything tried across this whole session in response to one observation: at `n_agents=10`, the CMA-ES-optimized `save_battery_avoid_all` controller burns far more battery than the fixed LJ baseline for essentially the same distance. All work below is the original architecture (10-input, no wall sensor), `n_agents=10`, seeded from each seed's own pre-existing `save_battery_avoid_wall` genome unless noted.

Current status: best validated result remains `HEBBIAN_EFF_DISTANCE_WEIGHT=16.0` alone (up from the original 8.0). Three follow-up attempts to reduce collision further (moderate collision penalty on top, a 2-stage curriculum, physical collision resolution) were all tried this round and all underperformed plain `weight=16` -- see the new section below. A weaker version of collision resolution is training now as the one still-open thread. `ants26_replication/experiment/config.py` currently has `HEBBIAN_RESOLVE_COLLISIONS=True` with the weaker (`strength=0.5`, `max_iter=2`) settings active for that in-progress run.

Correction (found late in the investigation): the LJ baseline's battery starts at 150 (its own native scale), while every trained genome's battery starts at 100. Every "baseline vs. trained" battery number in earlier drafts of this report compared raw units directly without normalizing — e.g. baseline "38.94" was compared to a genome's "17.42%" as if both were percentages of the same scale. **All baseline battery figures below are now normalized ($\div 150 \times 100$) to match the 0–100 scale everything else uses.** The gap this whole investigation has been chasing is real but smaller than earlier numbers implied — e.g. at `weight=16`, seed 123's genome (26.95%) already meets or slightly exceeds the corrected baseline (26.37%).

Investigation timeline

1 — OBSERVATION

Optimized controller burns notably more battery than the baseline, same distance

seed42: baseline covers 12.50m at 26.0% battery remaining (corrected); the trained genome covers 12.40m (statistically the same) but ends at 17.4%.

2 — DIAGNOSIS

Direct simulation revealed chronic inter-agent collision

The optimized swarm spends 195.5 of ~222s in active pairwise collision (spacing collapsing from 1.69 at spawn to 0.55 by episode end), vs. zero for baseline.

3 — FIX ATTEMPT: COLLISION PENALTY ALONE

collision_w 250 → 15: fixed collisions, made battery worse ABANDONED

Collision time dropped 195.5 → 30.5s, but battery went DOWN (17.4% → 14.8%). The tight, colliding formation was apparently providing some incidental battery benefit (likely wind-shielding/drafting) that was lost with nothing replacing it.

4 — FIX ATTEMPT: COHESION REWARD

Small reward for proximity, on top of the collision fix ABANDONED

`cohesion_w=2.0` worked great on seed42 (battery 22.2%) but FAILED on seeds 123 and 777 (one got worse on both battery and collision, the other's battery nearly halved). `cohesion_w=1.0` was even better on seed42 (22.7%, collision down to 6.5s) but was never validated on other seeds before being superseded by the distance-weight work below.

5 — ALTERNATIVE: HARDWARE-REALISTIC THYMIO II IR SENSOR

Built a whole new sensor variant — made things worse, not better

SEPARATE VARIANT, NOT PURSUED FURTHER

Replaced the idealized 2m-range sensor with a simulation of the real Thymio's 7-sensor, ~12cm-range IR array (angles sourced from Webots' calibrated model). Battery dropped to 9.0%, collision rose to 349.5s, plus a new wall-collision mode (1482) that didn't exist before. Confirms the idealized sensor was already the easiest possible setting for CMA-ES.

6 — MATLAB SOURCE COMPARISON

Found the actual reference implementation, line by line

Two concrete deviations found: (a) `KAPPA=20` in Python vs `kappa=10` in *both* MATLAB files, undocumented, unlike every other disclosed deviation in this codebase; (b) the 3-stage curriculum and the 8x distance weight do not exist anywhere in the MATLAB source — the reference trains under one unconditional formula (`dist + batt/5 - collision/250`, distance weight implicitly 1.0) from generation 1.

7 — FIX ATTEMPT: KAPPA=10

Correctness fix, but didn't touch the actual problem REVERTED

Halved wind drag increased every absolute number (baseline's own distance jumped 12.5 → 20m too), but the relative gap to baseline didn't close, and collision as a share of episode time got slightly worse (88% → 152%). Reverted to KAPPA=20 to keep results comparable with everything else in this report.

8 — DISTANCE WEIGHT SWEEP (4 / 8 / 16 / 32), 3 SEEDS EACH

The first genuinely consistent, multi-seed signal in the whole investigation

CURRENT BEST

4.0 was worse than 8.0 on every axis. 16.0 was clearly best: lowest average collision time (100.2s vs 8.0's 217.7s) across ALL 3 seeds, tied-best average battery (21.8%). 32.0 regressed most of the way back to 8.0's level (191.7s collision, 18.6% battery) — not a monotonic relationship, more like a local sweet spot near 16 that doubling overshoots.

9 — BATTERY-SCALE BUG FOUND, AND EVIDENCE THE ROTATION STRATEGY IS REAL

The baseline's battery is on a 0-150 scale, not 0-100 — every prior baseline comparison in this report was uncorrected

User asked to tune episode length so battery meaningfully constrains the baseline (target: ~75% used by natural termination). Testing directly against the baseline (no retraining needed, its control law doesn't reference battery at all) found the remaining-fraction is ALREADY ~29-30% regardless of battery budget (tested 1x/2x/4x) — a steady-state property, not something more simulation time fixes. That test surfaced the scale bug: baseline battery starts at 150 (`config.MAX_BATTERY`), not 100 like every trained genome. Corrected, baseline is really at ~26-35% remaining, not 39-52% — a real gap, but roughly half the size this report previously showed.

Separately: normalizing every genome's battery *standard deviation* onto the same 0-100 scale shows every single trained genome, at every distance weight tested, has meaningfully LOWER battery variance than the baseline (e.g. 10.5-17.6 vs. baseline's 20.8-24.8 percentage-points of std). That's the evolved "agents swap positions to share wind exposure" strategy showing up directly in the data -- it's real, and it's working, exactly as observed in the playback videos. It just isn't enough to overcome the separate, still-unsolved collision-waste problem: a fair split of a bigger total energy bill (collisions cost real battery with no offsetting benefit) still loses to an unfair split of a smaller one.

10 — THREE COLLISION-REDUCTION ATTEMPTS, ALL WORSE THAN PLAIN WEIGHT=16

Combining levers didn't compound -- each one alone underperformed the un-combined baseline

Three ideas tried in parallel (3 seeds each unless noted): **(a)** `weight=16 + collision_w=50` (moderate, between the original 250 and the earlier-tested-too-aggressive 15) -- battery dropped to ~11% across all 3 seeds (vs. `weight=16` alone's 15.7-27.0%). **(b)** 2-stage curriculum skipping `save_battery_avoid_wall` entirely, doubling `save_battery_avoid_all` to 200 generations to keep total budget constant -- also worse (seed123's collision count ballooned to 486.0, the worst number this whole investigation), suggesting stage 2 does real useful work rather than just entrenching bad habits for free. **(c)** physically preventing overlap (reviving a mechanism that exists, commented out, in the MATLAB reference's `move()`) -- n10_seed42 only: battery cratered to 7.06%, and the controller drove into MORE collision-course commands than anything else tried (357.5s), suggesting full-strength resolution removes the physical consequence of crowding so completely that nothing is left teaching the controller to avoid it.

11 — WEAKER COLLISION RESOLUTION (IN PROGRESS)

Soften the physical-resolution mechanism instead of abandoning it

Caught a design bug before wasting a run: naively lowering "strength" to 0.5 while keeping `max_iter=10` still converges to within ~0.001 of full resolution (each pass halves the remaining gap, so 10 halvings erases a 0.5-strength push just as completely as a 1.0-strength one) -- verified numerically, no real behavioral difference from attempt (c) above. Corrected version: `strength=0.5 AND max_iter=2`, which genuinely leaves residual crowding (~0.09m vs. the 0.12m floor) that persists into the next step instead of being erased. n10_seed42 training in progress.

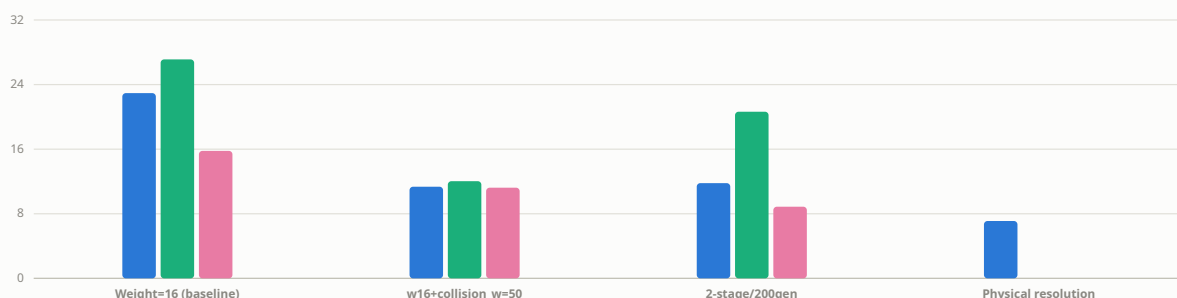
12 — COLLISION = INSTANT DEATH (QUEUED)

Make a full collision end the episode immediately, same pattern as battery-empty

New `HEBBIAN_COLLISION_INSTANT_DEATH` flag: the moment any pair crosses `min_dist`, the whole episode ends right there (mirroring the existing "any one agent's battery empties ends it for everyone" rule, applied to collision instead). Collision-count bookkeeping is unchanged, but a fatal collision now directly starves `dist_travelled` and `average_batt` by cutting the episode short, rather than relying on a fitness-penalty weight to be tuned just right. Mechanism smoke-tested (an untrained-for-this genome died after 27 steps/13.5s on first contact, as expected) and confirmed mutually exclusive with collision resolution (must run with `HEBBIAN_RESOLVE_COLLISIONS=False`, since resolution keeps every pair at `min_dist` by construction and would make this flag vacuous). Known risk, not yet observed: an early population may die almost instantly in nearly every candidate, giving CMA-ES too flat a landscape to climb -- worth watching the first several generations closely once launched. Queued to start after entry 11 finishes.

Collision-reduction attempts vs. the weight=16 baseline

Mean battery %, all three attempts vs. the unmodified weight=16 result at matching seeds. Resolution (full-strength) is seed 42 only.



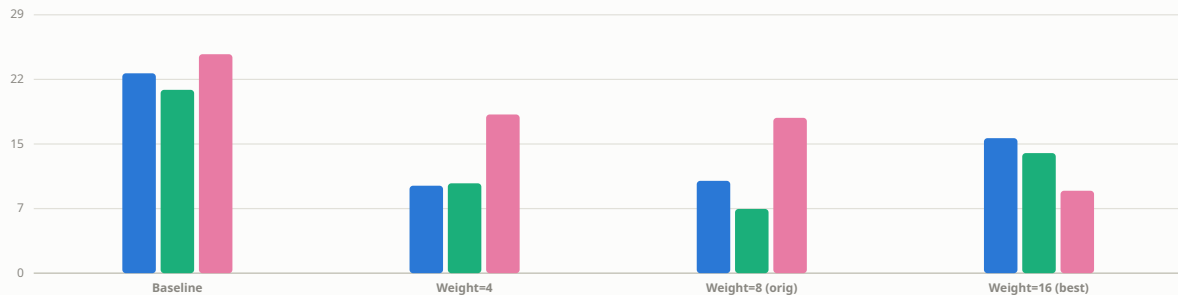
Every attempt underperforms plain weight=16 in every seed tested. None of the three ideas are being pursued further as originally specified; the weaker collision-resolution variant (entry 11

above) is the one open thread.

Battery variance: evidence the rotation strategy works

Standard deviation of final battery across agents, normalized to a 0-100 scale for all conditions.

Lower = more evenly distributed wind-exposure cost across the swarm.



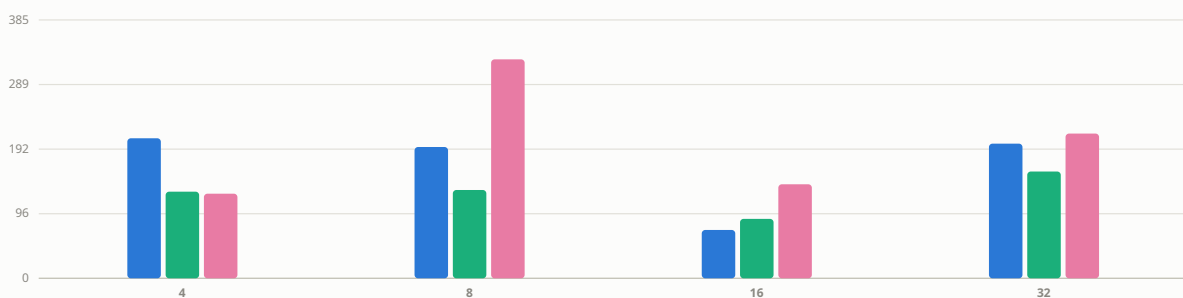
Every trained genome sits well below the baseline's variance, in every seed, at every distance weight tested (4, 8, and 16 all shown). The baseline's fixed potential-field rule has no fairness mechanism at all -- whichever agent ends up most wind-exposed just keeps paying for it. The evolved controller visibly does better here; the remaining gap in the mean is attributable to collision waste, not a failure to learn rotation.

Distance weight sweep — the core result

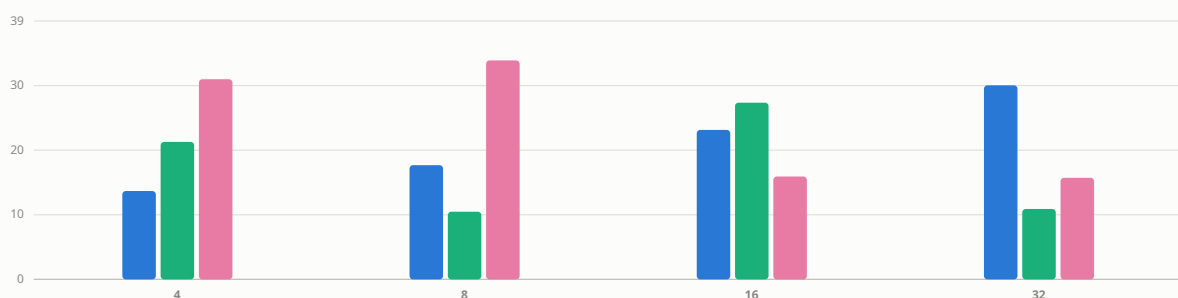
Direct simulation, same seed/n_agents/settings across all runs. 3 seeds per weight.

■ seed 42 ■ seed 123 ■ seed 777

Collision time (s) — lower is better

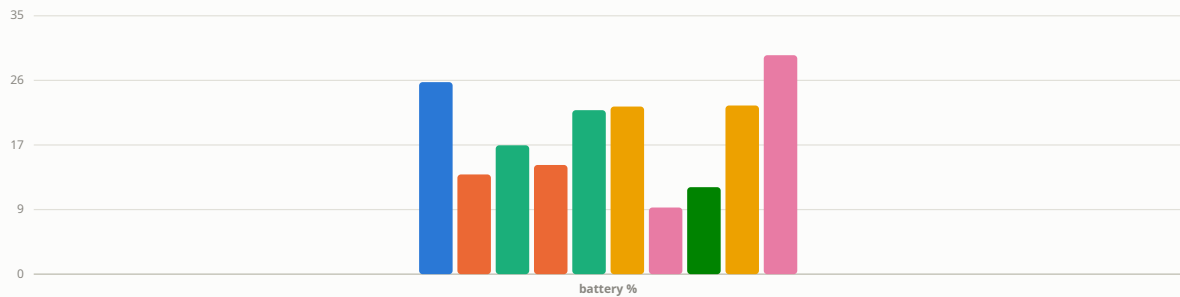


Mean final battery (%) — higher is better



Everything tried, seed 42 (the one constant reference point)

Every approach this session ran at n10_seed42 at some point, so this is the widest single comparison available. Not all bars are directly comparable in method (e.g. Thymio IR is a different sensor entirely) — see the timeline above for caveats.



Full sweep data

WEIGHT	SEED	DIST	BATTERY	BATT STD	COLLISION	EPISODE
4	42	13.1	13.5	9.9	208.5	134.5s
4	123	11.2	21.0	10.2	129.0	135.5s
4	777	8.3	30.5	18.0	126.0	107.5s
8	42	12.4	17.4	10.4	195.5	222.0s
8	123	11.8	10.3	7.3	131.5	143.0s
8	777	7.2	33.4	17.6	326.0	120.0s
16	42	11.0	22.8	15.3	72.0	108.0s
16	123	8.4	26.9	13.6	88.5	84.5s
16	777	9.7	15.7	9.3	140.0	148.5s
32	42	10.1	29.6	16.7	200.5	101.0s
32	123	13.0	10.7	7.0	159.0	146.5s
32	777	12.5	15.5	11.1	215.5	142.0s

Open questions / next steps

Fine-tune around 16. 8 → 16 helped, 16 → 32 hurt — the sweet spot might be somewhere in between or slightly past 16 (e.g. 12 or 20), not tested yet.

Weight=16 needs more seeds for real confidence. Every single-seed or 3-seed result this session has had at least one outlier (seed777 broke the cohesion_w=2.0 trend and partially broke the battery trend at weight=16 too). 3 seeds is better than 1, but still thin.

Even at the best result so far, the baseline is still ahead on average. Weight=16's average battery (21.8%) vs. baseline's corrected ~26-35% -- a real gap, though smaller than earlier numbers

in this report showed, and already closed on at least one seed (123).

Collision remains the main lever, but three targeted attempts at it all failed. The rotation/formation-sharing strategy is confirmed real (see the variance section above) -- what's missing is eliminating the collision-waste tax layered on top of it. Combining weight=16 with a moderate collision penalty, skipping stage 2 with double budget, and physically preventing overlap were all tried and all underperformed plain weight=16 (see the dedicated section above). A weaker version of physical resolution is the one thread still in progress.

The unstaged, literal-MATLAB-formula retrain is still untested -- and entry 10's 2-stage result (worse, not better, when stage 2 was removed) is a real data point suggesting the staged curriculum may be doing useful work, not just entrenching bad habits as originally hypothesized. Worth revisiting that assumption before trying the full unstaged formula.

File paths

WHAT	PATH
Best result so far (weight=16, all seeds)	hebbian_results_v2_original_distw16/n10_seed/ hebbian_save_battery_avoid_all_best.npy
Weight=4 sweep	hebbian_results_v2_original_distw4/n10_seed/
Weight=32 sweep	hebbian_results_v2_original_distw32/n10_seed/
Original (weight=8) sweep	hebbian_results_v2_original_sweep/n10_seed/
Collision-only fix	hebbian_results_v2_original_strongcollision/n10_seed42/
Cohesion w=2.0 (3 seeds)	hebbian_results_v2_original_cohesion/n10_seed/
Cohesion w=1.0 (seed42 only)	hebbian_results_v2_original_cohesion_w1/n10_seed42/
KAPPA=10 test	hebbian_results_v2_original_kappa10/n10_seed42/
Thymio II IR sensor variant	ants26_replication/thymio_ir_variant/, results in hebbian_results_v2_thymio_ir/n10_seed42/
Active config (currently weight=16)	ants26_replication/experiment/config.py
MATLAB reference source	ants-2026-polimi-energy-efficient-flocking-main/